Cryptography

Alessandro Vulcu

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Contents

1 Divisibility

Def 1.1. *Let $a, b \in \mathbb{Z}$ with $b \neq 0$. We say that b divides a, denoted by $b \mid a$, if there exists an integer $c \in \mathbb{Z}$ such that $a = b \cdot c$.*

Prop 1.1: Transitivity of Divisibility. *Let $a, b, c \in \mathbb{Z}$. If $a \mid b$ and $b \mid c$, $\rightarrow a \mid c$.*

Proof. Assume $a \mid b$. Since $a, b \in \mathbb{Z}$ and $a \neq 0$, $\exists c_1 \in \mathbb{Z} : b = a \cdot c_1$. Assume $b \mid c$. Since $b, c \in \mathbb{Z}$ and $b \neq 0$, $\exists c_2 \in \mathbb{Z} : c = b \cdot c_2$.

Substituting the first equation into the second yields:

$$c = (a \cdot c_1) \cdot c_2 = a \cdot (c_1 c_2)$$

Since $c_1, c_2 \in \mathbb{Z}$, their product $c_1 c_2$ is also an integer. Let $k = c_1 c_2 \in \mathbb{Z}$. Thus, $c = a \cdot k$. By the definition of divisibility, we conclude that $a \mid c$. □

Prop 1.2: Equality from Mutual Divisibility. *Let $a, b \in \mathbb{Z}$, with $a \neq 0$ and $b \neq 0$. If $a \mid b$ and $b \mid a$, $\rightarrow a = \pm b$.*

Proof. Assume $a \mid b$. Since $a \neq 0$, $\exists c_1 \in \mathbb{Z} : b = a \cdot c_1$. Assume $b \mid a$. Since $b \neq 0$, $\exists c_2 \in \mathbb{Z} : a = b \cdot c_2$.

Substitute the expression for b from the first equation into the second:

$$a = (a \cdot c_1) \cdot c_2 = a \cdot (c_1 c_2)$$

Since $a \neq 0$, we can divide both sides by a, which gives:

$$1 = c_1 c_2$$

Since c_1 and c_2 are integers, the only possible pairs for (c_1, c_2) are $(1, 1)$ or $(-1, -1)$.

Case 1: $c_1 = 1$. Substituting into $b = a \cdot c_1$ gives $b = a \cdot 1 = a$. Case 2: $c_1 = -1$. Substituting into $b = a \cdot c_1$ gives $b = a \cdot (-1) = -a$.

Therefore, we conclude that $a = \pm b$. □

Prop 1.3: Linearity Property of Divisibility. *Let $a, b, c \in \mathbb{Z}$, with $a \neq 0$. If $a \mid b$ and $a \mid c$, $\rightarrow a \mid (b + c)$ and $a \mid (b - c)$.*

Proof. Assume $a \mid b$. Since $a \neq 0$, $\exists k_1 \in \mathbb{Z} : b = a \cdot k_1$. Assume $a \mid c$. Since $a \neq 0$, $\exists k_2 \in \mathbb{Z} : c = a \cdot k_2$.

Demonstration for the Sum ($a \mid (b + c)$): Summing the two equations:

$$b + c = a \cdot k_1 + a \cdot k_2$$

Factoring out a:

$$b + c = a \cdot (k_1 + k_2)$$

Since $k_1, k_2 \in \mathbb{Z}$, their sum (k_1+k_2) is also an integer. Let $k_s = k_1+k_2 \in \mathbb{Z}$. Thus, $b+c = a \cdot k_s$. By the definition of divisibility, we have $a \mid (b + c)$.

Demonstration for the Difference ($a \mid (b-c)$): Subtracting the second equation from the first:

$$b - c = a \cdot k_1 - a \cdot k_2$$

Factoring out a:

$$b - c = a \cdot (k_1 - k_2)$$

Since $k_1, k_2 \in \mathbb{Z}$, their difference $(k_1 - k_2)$ is also an integer. Let $k_d = k_1 - k_2 \in \mathbb{Z}$. Thus, $b - c = a \cdot k_d$. By the definition of divisibility, we have $a \mid (b - c)$. □

Prop 1.4: Euclidean Division. *Let $a \in \mathbb{N}$ and $b \in \mathbb{N}$ with $b > 0$. Then $\exists! q, r \in \mathbb{Z} : a = bq + r$ with $0 \leq r < b$.*

Prop 1.5: Binary Representation from Repeated Division. *Let $m \in \mathbb{N}$, with $m < 2^B$. The repeated division-by-2 process:*

$$\begin{aligned} m &= 2q_0 + m_0 \\ q_0 &= 2q_1 + m_1 \\ q_1 &= 2q_2 + m_2 \\ &\cdots \end{aligned}$$

terminates with $q_{B-1} = 0$, and $m = m_0 + 2m_1 + \cdots + 2^{B-1}m_{B-1}$.

Proof. The proof consists of two parts: demonstrating the polynomial form and proving that the process terminates.

Polynomial Form

We can show the relationship by algebraic substitution.

$$m = 2q_0 + m_0$$

Substitute $q_0 = 2q_1 + m_1$:

$$m = 2(2q_1 + m_1) + m_0 = 2^2 q_1 + 2m_1 + m_0$$

Substitute $q_1 = 2q_2 + m_2$:

$$m = 2^2(2q_2 + m_2) + 2m_1 + m_0 = 2^3 q_2 + 2^2 m_2 + 2m_1 + m_0$$

Continuing this process until the final step where $q_{B-2} = 2q_{B-1} + m_{B-1}$, we arrive at the final form: $m = m_0 + 2m_1 + 2^2 m_2 + \cdots + 2^{B-1} m_{B-1}$.

Termination

The process is guaranteed to terminate because the sequence of quotients is strictly decreasing and bounded.

- Since $m < 2^B$, the first division $m = 2q_0 + m_0$ implies $2q_0 \leq m < 2^B$, so $q_0 < 2^{B-1}$.
- Similarly, $q_0 < 2^{B-1}$ implies $2q_1 \leq q_0 < 2^{B-1}$, so $q_1 < 2^{B-2}$.

This pattern continues: $q_i < 2^{B-1-i}$. For the final quotient q_{B-1}, we have $q_{B-1} < 2^{B-1-(B-1)} = 2^0 = 1$. Since $q_{B-1} \in \mathbb{N} \cup \{0\}$, the only possibility is $q_{B-1} = 0$. The process terminates when the quotient becomes zero. □

Rem: Upper Bound of a B-bit Number. *If m has B bits in its binary representation, i.e., $m = (m_{B-1} \dots m_0)_2$, then $m < 2^B$.*

Proof. The largest possible value for a B-bit number occurs when all bits are 1. This value can be calculated by summing the geometric series:

$$m \leq \sum_{i=0}^{B-1} 2^i = \frac{2^B - 1}{2 - 1} = 2^B - 1$$

Therefore, $m < 2^B$. □

Cor 1.1: Range of a B-bit Number. *A natural number $m \in \mathbb{N}$ has a binary representation with B bits (where the most significant bit $m_{B-1} = 1$) $\iff$ $2^{B-1} \leq m < 2^B$.*

2 The Greatest Common Divisor (GCD)

Def 2.1: Greatest Common Divisor. *Let $a, b \in \mathbb{Z}$, such that $a \neq 0$ or $b \neq 0$. The set of their common divisors has a largest element $d \in \mathbb{N}$, called the greatest common divisor of a and b. We write $d = \gcd(a, b)$.*

Rem: Example of GCD. *To find $\gcd(18, 12)$, we can list the divisors of each number:*

- *Divisors of 12:* $\{1, 2, 3, 4, 6, 12\}$
- *Divisors of 18:* $\{1, 2, 3, 6, 9, 18\}$

The set of common divisors is $\{1, 2, 3, 6\}$. The largest element in this set is 6, so $\gcd(18, 12) = 6$.

While listing all divisors is feasible for small numbers, it is impractical for large integers. A far more powerful and efficient method is the Euclidean Algorithm.

3 The Euclidean Algorithm

Th 3.1: Euclidean Algorithm. *Let $a, b \in \mathbb{N} : a > 0, b > 0$ and $a \geq b$. The following algorithm computes $\gcd(a, b)$ in a finite number of steps:*

1. *Let $r_0 := a$ and $r_1 := b$.*
2. *Set $i = 1$.*
3. *Divide r_{i-1} by r_i to obtain a quotient q_i and remainder r_{i+1}: $r_{i-1} = r_i q_i + r_{i+1}$.*
4. *If $r_{i+1} = 0$, then $r_i = \gcd(a, b)$. The algorithm terminates.*
5. *Otherwise, increment $i \to i + 1$ and return to Step 3.*

Proof. Let the sequence of remainders be $r_0, r_1, \ldots, r_n, r_{n+1}$ where $d = r_n$ is the last non-zero remainder and $r_{n+1} = 0$. We must demonstrate two facts: that d is a common divisor of a and b, and that it is the greatest common divisor.

a. $d \mid a$ and $d \mid b$

We proceed by working backward from the end of the algorithm.

- The final division step is $r_{n-1} = r_n q_n + r_{n+1}$. Since $r_{n+1} = 0$ and $d = r_n$, this equation becomes $r_{n-1} = dq_n$. This demonstrates $d \mid r_{n-1}$.
- The preceding step is $r_{n-2} = r_{n-1} q_{n-1} + r_n$. Since $d \mid r_n$ and we have shown $d \mid r_{n-1}$, it follows that d must divide their linear combination, $d \mid r_{n-2}$.

Continuing this backward induction process, we establish that d divides every preceding remainder. This concludes that $d \mid r_1$ (where $r_1 = b$) and $d \mid r_0$ (where $r_0 = a$). Thus, d is a common divisor.

b. d is the Greatest Common Divisor

Let $c \in \mathbb{Z}$ be any common divisor of a and b. We work forward down the chain of remainders.

- Since $c \mid a$ (r_0) and $c \mid b$ (r_1), and $r_2 = r_0 - r_1 q_1$, it must be that $c \mid r_2$.
- Since $c \mid r_1$ and $c \mid r_2$, and $r_3 = r_1 - r_2 q_2$, it must be that $c \mid r_3$.

Continuing this process, we eventually find that c must divide the last non-zero remainder, r_n, so $c \mid d$. Since $c \mid d$, it is a consequence of divisibility that $c \leq d$. This holds $\forall c$ common divisor, proving that d is the greatest common divisor. □

Prop 3.1: Extended Euclidean Algorithm. *Let $a, b \in \mathbb{N}$ such that $a \neq 0$ or $b \neq 0$. If $d = \gcd(a, b)$, then $\exists u, v \in \mathbb{Z} : d = au + bv$.*

Proof. The proof relies on the standard Euclidean Algorithm applied to a and b, which generates a sequence of remainders r_i:

$$\begin{aligned}
a &= q_1 b + r_1, \quad \text{with } 0 \leq r_1 < b \\
b &= q_2 r_1 + r_2, \quad \text{with } 0 \leq r_2 < r_1 \\
r_1 &= q_3 r_2 + r_3, \quad \text{with } 0 \leq r_3 < r_2 \\
&\vdots \\
r_{n-3} &= q_{n-1} r_{n-2} + r_{n-1}, \quad \text{with } 0 \leq r_{n-1} < r_{n-2} \\
r_{n-2} &= q_n r_{n-1} + 0.
\end{aligned}$$

Since r_{n-1} is the last non-zero remainder, we have $d = \gcd(a, b) = r_{n-1}$.

We proceed by **backward substitution** (as indicated by the notes on the right):

- From the second-to-last equation, r_{n-1} can be expressed as a linear combination of r_{n-3} and r_{n-2}:
$$d = r_{n-1} = r_{n-3} - q_{n-1} r_{n-2}$$
- Since r_{n-2} is a linear combination of r_{n-4} and r_{n-3}, substituting this expression for r_{n-2} into the equation for d will express d as a linear combination of r_{n-4} and r_{n-3}.
- By continuing this backward substitution process (i.e., replacing each remainder r_i with its expression from the previous division step in the Euclidean algorithm), we progressively express d as a linear combination of the previous two terms in the remainder sequence:
$$r_i = \text{lin. comb. of } r_{i-2} \text{ and } r_{i-1}$$
- This process continues until we reach the first two equations, where d will be expressed as a linear combination of a and b:
$$d = r_{n-1} = au + bv$$
for some integers $u, v \in \mathbb{Z}$.

□

Prop 3.2: Solvability of Linear Diophantine Equations. *Let $a, b \in \mathbb{Z}$ be integers, not both zero. The equation $au + bv = c$ has integer solutions $u, v \in \mathbb{Z}$ if and only if* $\gcd(a, b) \mid c$.

Proof. ($\Longleftarrow$) **If** $\gcd(a, b) \mid c$**, solutions exist**
Assume $c = k \cdot \gcd(a, b)$ for some $k \in \mathbb{Z}$. By Bézout's Identity, $\exists u_0, v_0 \in \mathbb{Z}$ so that $au_0 + bv_0 = \gcd(a, b)$.

Multiplying by k yields:

$$k(au_0 + bv_0) = k \cdot \gcd(a, b) \implies a(ku_0) + b(kv_0) = c$$

Thus, $u = ku_0$ and $v = kv_0$ are integer solutions.

($\Longrightarrow$) **If solutions exist, then** $\gcd(a, b) \mid c$
Assume $au + bv = c$ has integer solutions $u, v \in \mathbb{Z}$.

Let $d = \gcd(a, b)$. Since $d \mid a$ and $d \mid b$, d must divide any linear combination of a and b:

$$d \mid (au + bv) \implies d \mid c$$

□

Prop 3.3: All Integer Solutions to $au + bv = c$. *Let $a, b \in \mathbb{Z}$ be relatively prime (i.e.,* $\gcd(a, b) = 1$*), and let $u_0, v_0 \in \mathbb{Z}$ be a particular solution so that $au_0 + bv_0 = c$. Then, all integer solutions (u, v) to $au + bv = c$ are of the form:*

$$u = u_0 + kb \quad \textit{and} \quad v = v_0 - ka$$

for any $k \in \mathbb{Z}$.

Proof. **Part 1: Verification (The form gives solutions)**
Substituting the general form into the equation:

$$\begin{aligned} a(u_0 + kb) + b(v_0 - ka) &= au_0 + akb + bv_0 - bka \\ &= (au_0 + bv_0) + (akb - bka) \\ &= c + 0 \\ &= c \end{aligned}$$

Thus, $\forall k \in \mathbb{Z}$, the formula yields a solution.

Part 2: Completeness (All solutions must be of this form)
Assume (u, v) is any arbitrary solution, so $au + bv = c$. We also have the particular solution $au_0 + bv_0 = c$.

Subtracting the two equations gives:

$$a(u - u_0) + b(v - v_0) = 0 \implies a(u - u_0) = -b(v - v_0) \implies a(u - u_0) = b(v_0 - v)$$

Since $a \mid b(v_0 - v)$ and we are given $\gcd(a, b) = 1$, by Euclid's Lemma, it must be that a divides the other factor, $(v_0 - v)$:

$$a \mid (v_0 - v)$$

Let $k \in \mathbb{Z}$ be the integer so that $v_0 - v = ka$. Rearranging this gives the form for v:

$$v = v_0 - ka$$

Substituting $v_0 - v = ka$ back into the equality $a(u - u_0) = b(v_0 - v)$:

$$a(u - u_0) = b(ka) = a(kb)$$

Since $a \neq 0$, we can divide by a to get:

$$u - u_0 = kb$$

Rearranging this gives the form for u:

$$u = u_0 + kb$$

Hence, all solutions are of the specified form. $\square$

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4 Coprime numbers

Def 4.1: Prime Number. *A natural number $p \in \mathbb{N}$ such that $p \geq 2$ is called **prime** if $\forall$ divisors $d \in \mathbb{N}$ of p, we have $d = 1$ or $d = p$.*

Th 4.1: Euclid. *There are infinitely many prime numbers.*

Proof. We proceed by contradiction.

Assume the contrary: that the set of all prime numbers is finite.

Let $N \in \mathbb{N}$ be the largest prime number so that $\forall$ prime number p, we have $p \leq N$.

Consider the natural number $M = N! + 1$.

Since $N \geq 2$, we have $M = N! + 1 > N$. Since M is greater than the largest assumed prime N, it follows that M **is not prime**.

Therefore, M must have a prime divisor. Let $p \in \mathbb{N}$ be any prime divisor of M.

Since p is a prime number and we assumed N is the largest prime, it must be that $\forall$ prime divisor p of M, we have $p \leq N$.

Since $p \leq N$, it follows that p is one of the factors in the factorial $N! = 1 \cdot 2 \cdot 3 \cdots N$.

$$\implies p \mid N!$$

Since we defined p as a divisor of M, we also have:

$$\implies p \mid (N! + 1)$$

Since p divides both $N! + 1$ and $N!$, p must divide their difference:

$$p \mid ((N! + 1) - N!)$$
$$p \mid 1$$

This is a **contradiction**, because a prime number $p \geq 2$ cannot divide 1.

Therefore, the initial assumption that there exists a largest prime N must be false.

Thus, there must exist a prime number $p > N$. Since N was assumed to be the largest prime, the existence of a larger prime p proves that the set of prime numbers is **infinite**. □

Prop 4.1: Divisors of Composite Numbers. *If a natural number $n \in \mathbb{N}$ is not prime (i.e., composite), then n has a divisor $a \in \mathbb{N}$ such that $1 < a \leq \sqrt{n}$.*

Proof. If n is not prime (composite), then n must be the product of two integer factors a and b, both greater than 1. So, there exist $a, b \in \mathbb{N}$ such that $a > 1$, $b > 1$, and $n = ab$.

We proceed by contradiction on the size of the factors. Assume the contrary: that $\forall$ factors a and b of n, we have both $a > \sqrt{n}$ and $b > \sqrt{n}$.

If $a > \sqrt{n}$ and $b > \sqrt{n}$, then multiplying these two inequalities yields:

$$\begin{aligned} n = ab &> (\sqrt{n}) \cdot (\sqrt{n}) \\ n &> (\sqrt{n})^2 \\ n &> n \end{aligned}$$

This is a **contradiction**, as n cannot be strictly greater than itself.

Therefore, the assumption that both factors a and b are greater than $\sqrt{n}$ must be false. This means that at least one of the factors must be less than or equal to $\sqrt{n}$.

Let a be the smaller of the two factors. We have $1 < a$ (by definition of a composite number) and $a \leq \sqrt{n}$. Thus, n has a divisor a such that $1 < a \leq \sqrt{n}$. □

Prop 4.2: Characterisation of Coprime Numbers. *Integers $a, b \in \mathbb{Z}$ are relatively prime $\iff \exists u, v \in \mathbb{Z} : au + bv = 1$.*

Proof.

- ($\Rightarrow$) This is a direct consequence of the Extended Euclidean Algorithm. If $\gcd(a, b) = 1$, then the algorithm guarantees $\exists u, v \in \mathbb{Z} : au + bv = 1$.
- ($\Leftarrow$) Conversely, assume $\exists u, v \in \mathbb{Z} : au + bv = 1$. Let $d = \gcd(a, b)$. Since $d \mid a$ and $d \mid b$, it must divide their linear combination $au + bv$. Therefore, $d \mid 1$. The only $d \in \mathbb{N}$ so that $d \mid 1$ is $d = 1$. □

Cor 4.1. *Let p be a prime number and $m \in \mathbb{Z}$. If $p \nmid m$, then $\gcd(p, m) = 1$.*

Proof. The positive divisors of a prime number p are only $\{1, p\}$. If $p \nmid m$, then p is not a common divisor. Therefore, the only common divisor of p and m is 1, which means $\gcd(p, m) = 1$. □

Prop 4.3: Prime Divisor Property. *Let p be a prime and $a, b \in \mathbb{Z}$. If $p \mid ab$, then $p \mid a$ or $p \mid b$.*

Proof. Assume $p \mid ab$. If $p \mid a$, the proposition holds. If $p \nmid a$, then by the preceding corollary, $\gcd(p, a) = 1$. By the characterization of coprime numbers, $\exists u, v \in \mathbb{Z} : pu + av = 1$. Multiplying this equation by b gives $pub + abv = b$.

Since $p \mid pub$ and we are given $p \mid ab$, which implies $p \mid abv$. Since p divides both terms on the left side, it must divide their sum. Therefore, $p \mid (pub + abv) \to p \mid b$. □

Th 4.2: Fundamental Theorem of Arithmetic. *Let $a \in \mathbb{N}$ so that $a \geq 2$. Then a can be factored as a product of primes, $a = p_1^{e_1} p_2^{e_2} \cdots p_m^{e_m}$, where p_i are distinct primes, and $\forall i \in \{1, \ldots, m\}, e_i \geq 1$. The factorization into primes is unique modulo rearrangements of the order of the factors.*

Rem. *The exponent $e_i = mult_{p_i}(a)$ is the multiplicity of p_i in a. A generalized form of the prime factorization is given by:*

$$a = \prod_{p \text{ prime}} p^{mult_p(a)}$$

where $mult_p(a) := 0$ if p is not a prime factor of a.

5 Classes Modulo m

Prop 5.1. *$\forall a \in \mathbb{Z}$ (the dividend) and $\forall m \in \mathbb{N}$ so that $m > 1$ (the modulus), $\exists! q, r \in \mathbb{Z} : a = mq + r$, with $0 \leq r < m$.*

Th 5.1: Existence of a Unique Remainder. *$\forall a \in \mathbb{Z}$ and $\forall m \in \mathbb{N} : m > 1$, $\exists! r \in \{0, 1, \ldots, m-1\} : a \equiv r \pmod{m}$.*

Proof. By Proposition 5.1 (Euclidean Division), we have $\exists! q, r \in \mathbb{Z} : a = mq + r$, where $0 \leq r < m$. Rearranging this equation gives $a - r = mq$. By the formal definition of divisibility, $m \mid (a - r)$. By the formal definition of congruence, this directly implies $a \equiv r \pmod{m}$. The uniqueness of r is guaranteed by the Euclidean Division Theorem. □

Th 5.2: Congruence is an Equivalence Relation. *The relation of congruence modulo m (denoted by $\equiv \pmod{m}$) is an equivalence relation on the set of integers $\mathbb{Z}$.*

Proof. We must demonstrate that congruence modulo m satisfies reflexivity, symmetry, and transitivity.

- **Reflexivity:** $\forall a \in \mathbb{Z}$, $a - a = 0$. Since $m \mid 0$, $a \equiv a \pmod{m}$. The relation is reflexive.
- **Symmetry:** Assume $a \equiv b \pmod{m}$. $\implies m \mid (a - b)$, so $\exists k \in \mathbb{Z} : a - b = km$. Multiplying by -1 yields $b - a = (-k)m$. Since $-k \in \mathbb{Z}$, $\implies m \mid (b - a)$, so $b \equiv a \pmod{m}$. The relation is symmetric.
- **Transitivity:** Assume $a \equiv b \pmod{m}$ and $b \equiv c \pmod{m}$. $\implies \exists k_1, k_2 \in \mathbb{Z} : a - b = k_1 m$ and $b - c = k_2 m$. Adding the equations:

  $$(a - b) + (b - c) = k_1 m + k_2 m$$

  $$a - c = (k_1 + k_2)m$$

  Since $k_1 + k_2 \in \mathbb{Z}$, $\implies m \mid (a - c)$, so $a \equiv c \pmod{m}$. The relation is transitive. □

Prop 5.2: Equivalence Classes Form a Partition. *Let $\sim$ be an equivalence relation on a set X. The set of equivalence classes, $X/\sim$, forms a partition of X.*

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6 Invertibility Modulo m

Prop 6.1: Compatibility. *Let* $a_1, a_2, b_1, b_2 \in \mathbb{Z}$. *If* $a_1 \equiv a_2 \pmod{m}$ *and* $b_1 \equiv b_2 \pmod{m}$, *then:*

1. $a_1 + b_1 \equiv a_2 + b_2 \pmod{m}$
2. $a_1 b_1 \equiv a_2 b_2 \pmod{m}$

Proof.

- **Addition:** Given $a_2 \equiv a_1 \pmod{m}$ and $b_2 \equiv b_1 \pmod{m}$, $\implies \exists k_1, k_2 \in \mathbb{Z} : a_2 - a_1 = k_1 m$ and $b_2 - b_1 = k_2 m$. Adding the equations yields $(a_2 - a_1) + (b_2 - b_1) = (k_1 + k_2)m$. Rearranging the terms: $(a_2 + b_2) - (a_1 + b_1) = (k_1 + k_2)m$. Since $k_1 + k_2 \in \mathbb{Z}$, this shows $m \mid ((a_2 + b_2) - (a_1 + b_1))$, so $a_1 + b_1 \equiv a_2 + b_2 \pmod{m}$.
- **Multiplication:** We express $a_2 = a_1 + k_1 m$ and $b_2 = b_1 + k_2 m$. Multiplying them gives:

  $$a_2 b_2 = (a_1 + k_1 m)(b_1 + k_2 m) = a_1 b_1 + a_1 k_2 m + b_1 k_1 m + k_1 k_2 m^2$$

  Factoring out m: $a_2 b_2 - a_1 b_1 = m(a_1 k_2 + b_1 k_1 + k_1 k_2 m)$. Let $K = a_1 k_2 + b_1 k_1 + k_1 k_2 m$. Since $K \in \mathbb{Z}$, $m \mid (a_2 b_2 - a_1 b_1)$, so $a_1 b_1 \equiv a_2 b_2 \pmod{m}$. □

Prop 6.2. *The set of integers modulo m under addition, $(\mathbb{Z}/m\mathbb{Z}, +)$, forms a* ***commutative group****.*

Prop 6.3. *When considering both addition and multiplication, $(\mathbb{Z}/m\mathbb{Z}, +, \cdot)$ forms a* ***commutative ring****.*

Th 6.1. *If an inverse exists, it is unique.*

Proof. Assume an element $[a]$ has two inverses, $[a']$ and $[a'']$. By definition, $[a] \cdot [a'] = [1]$ and $[a] \cdot [a''] = [1]$. Since multiplication is associative and $[1]$ is the identity element, we can write:

$$[a'] = [a'] \cdot [1] = [a'] \cdot ([a] \cdot [a''])$$

$$[a'] = ([a'] \cdot [a]) \cdot [a''] = [1] \cdot [a''] = [a'']$$

Therefore, $[a'] = [a'']$, $\implies$ the inverse is unique. □

Th 6.2: Characterization of Invertibility. *An element $[a] \in \mathbb{Z}/m\mathbb{Z}$ is invertible $\iff$ $\gcd(a,m) = 1$.*

Proof. An element $[a]$ is invertible $\iff$ $\exists u \in \mathbb{Z} : au \equiv 1 \pmod{m}$. This congruence is equivalent to the linear Diophantine equation $au - 1 = -mv$ for some $v \in \mathbb{Z}$, which can be rewritten as $au + mv = 1$. By Bézout's Identity, this equation has an integer solution $\iff$ $\gcd(a,m) \mid 1$. Since $\gcd(a,m) \in \mathbb{N}$, this condition is satisfied $\iff$ $\gcd(a,m) = 1$. □

Prop 6.4. *The set of units under multiplication, $(\mathbb{Z}/m\mathbb{Z})^*$, forms a* ***commutative group*** *(the group of units modulo m).*

Prop 6.5. *$\forall p$ prime number, the commutative ring $(\mathbb{Z}/p\mathbb{Z}, +, \cdot)$ is a* ***field****, commonly denoted $\mathbb{F}_p$.*

Cor 6.1. *Let $p \in \mathbb{N}$ be a prime so that $p > 2$. The equation $x^2 \equiv 1 \pmod{p}$ in $\mathbb{F}_p$ has exactly two distinct solutions: $x \equiv 1 \pmod{p}$ and $x \equiv p - 1 \pmod{p}$ (i.e., $x \equiv -1 \pmod{p}$).*

Proof. The equation $x^2 \equiv 1 \pmod{p}$ is equivalent to $p \mid (x^2 - 1)$, or $p \mid (x-1)(x+1)$. Since p is prime, by the Prime Divisor Property:

- Either $p \mid (x - 1)$, which implies $x \equiv 1 \pmod{p}$.
- Or $p \mid (x + 1)$, which implies $x \equiv -1 \equiv p - 1 \pmod{p}$.

These two solutions are distinct because $1 \not\equiv -1 \pmod{p}$ for $p > 2$. □

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7 Solving Linear Congruences

Prop 7.1: Solvability Condition. *The linear congruence* $ax \equiv b \pmod{m}$ *has at least one integer solution for* $x \in \mathbb{Z} \iff d = \gcd(a, m) \mid b$.

Proof. The congruence $ax \equiv b \pmod{m}$ is equivalent to the linear Diophantine equation $ax + mv = b$ for some $v \in \mathbb{Z}$. This equation is solvable for $x, v \in \mathbb{Z} \iff$ the greatest common divisor of the coefficients, $d = \gcd(a, m)$, divides the constant term b. This follows directly from the Solvability Condition for Diophantine Equations (Bézout's identity). $\square$

Prop 7.2: Number of Solutions. *If* $d = \gcd(a, m) \mid b$*, the congruence* $ax \equiv b \pmod{m}$ *has exactly* d *distinct solutions modulo* m.

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8 Chinese Remainder Theorem (CRT)

Chinese Remainder Theorem (CRT)

Th 8.1: Chinese Remainder Theorem. *Let $m_1, m_2, \dots, m_k \in \mathbb{N}$ be pairwise relatively prime integers (i.e., $\forall i \neq j, \gcd(m_i, m_j) = 1$). Let $a_1, a_2, \dots, a_k \in \mathbb{Z}$ be arbitrary integers. The system of congruences:*

$$x \equiv a_i \pmod{m_i} \quad \forall i \in \{1, \dots, k\}$$

has a solution, and this solution is unique modulo $M = \prod_{i=1}^{k} m_i$.

Proof. The proof is constructive for existence and uses coprimality for uniqueness. Let $M = \prod_{i=1}^{k} m_i$ and $N_j = M/m_j$.

Existence (Construction)

Since $\gcd(N_j, m_j) = 1$, $\exists v_j \in \mathbb{Z}$ so that $v_j N_j \equiv 1 \pmod{m_j}$. We construct the solution x:

$$x = \sum_{j=1}^{k} a_j v_j N_j$$

For an arbitrary index i, consider $x \pmod{m_i}$. $\forall j \neq i$, $m_i \mid N_j \implies N_j \equiv 0 \pmod{m_i}$. Thus, all terms $j \neq i$ vanish modulo m_i:

$$x \equiv a_i v_i N_i + \sum_{j \neq i} (0) \equiv a_i(1) \equiv a_i \pmod{m_i}$$

A solution x exists.

Uniqueness (Modulo M)

Assume x and x' are two solutions. Since $x \equiv a_i \pmod{m_i}$ and $x' \equiv a_i \pmod{m_i}$ for all i, we have $x - x' \equiv 0 \pmod{m_i}$, meaning $m_i \mid (x - x')\ \forall i$. Since the m_i are pairwise coprime, their product $M = \prod m_i$ must also divide the difference:

$$M \mid (x - x') \implies x \equiv x' \pmod{M}$$

The solution is unique modulo M. $\square$

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9 Euler's ϕ Function

Def 9.1: Euler's Totient Function, ϕ. *Let $m \in \mathbb{N}$ such that $m \geq 1$. The **Euler's totient function**, denoted by $\phi(m)$, is the number of elements in the multiplicative group of integers modulo m, which is $(\mathbb{Z}/m\mathbb{Z})^*$.*

In other words, $\phi(m)$ is the number of integers k such that $1 \leq k \leq m$ so that $\gcd(k, m) = 1$.

Def 9.2: Remainder Map. *Let $m, n \in \mathbb{N}$ be integers. The **remainder map** r is a function:*

$$r : \mathbb{Z}/(mn)\mathbb{Z} \longrightarrow (\mathbb{Z}/m\mathbb{Z}) \times (\mathbb{Z}/n\mathbb{Z})$$

It is defined $\forall a \in \{0, \ldots, mn - 1\}$ as:

$$r([a]_{mn}) := ([a]_m, [a]_n)$$

where $[a]_k$ denotes the residue class of a modulo k.

Prop 9.1: Remainder Map is Well-Defined. *The remainder map r is well-defined. That is, the result of the map is independent of the choice of the representative of the residue class:*

$$[a]_{mn} = [b]_{mn} \implies r([a]_{mn}) = r([b]_{mn})$$

which means $([a]_m, [a]_n) = ([b]_m, [b]_n)$.

Proof. We assume the antecedent: $[a]_{mn} = [b]_{mn}$.

By the definition of equality of residue classes, this means mn divides the difference $(b - a)$:

$$[a]_{mn} = [b]_{mn} \implies mn \mid (b - a)$$

Since $mn \mid (b - a)$, it must be that both m and n individually divide the difference $(b - a)$:

$$mn \mid (b - a) \implies \begin{cases} m \mid (b - a) \\ n \mid (b - a) \end{cases}$$

By the definition of congruence and residue classes, this translates back into the desired equalities:

$$\begin{cases} m \mid (b - a) \implies [a]_m = [b]_m \\ n \mid (b - a) \implies [a]_n = [b]_n \end{cases}$$

Since both components are equal, the ordered pairs are equal:

$$([a]_m, [a]_n) = ([b]_m, [b]_n)$$

Thus, $r([a]_{mn}) = r([b]_{mn})$, and the map is well-defined. □

Th 9.1: Bijectivity of the Remainder Map with Coprime Factors. *Let $m, n \in \mathbb{N}$ be integers such that $m, n \geq 1$, and assume m and n are **pairwise coprime** (i.e., $\gcd(m, n) = 1$).*

The remainder map r:

$$r : \mathbb{Z}/(mn)\mathbb{Z} \longrightarrow (\mathbb{Z}/m\mathbb{Z}) \times (\mathbb{Z}/n\mathbb{Z})$$

*is a **bijection** (one-to-one and onto).*

Proof. The proof establishes bijectivity by demonstrating both **Surjectivity** (the map is onto) and **Injectivity** (the map is one-to-one).

Surjectivity (Onto)

Let $([a]_m, [b]_n) \in (\mathbb{Z}/m\mathbb{Z}) \times (\mathbb{Z}/n\mathbb{Z})$ be an arbitrary element in the codomain.

We seek to show $\exists x \in \mathbb{Z}$ so that $r([x]_{mn}) = ([a]_m, [b]_n)$.

This is equivalent to finding an integer x that simultaneously solves the system of congruences:

$$\begin{cases} x \equiv a \pmod{m} \\ x \equiv b \pmod{n} \end{cases}$$

Since we are given that $\gcd(m, n) = 1$, the **Chinese Remainder Theorem** guarantees that a solution x always exists for any pair of residues (a, b).

Therefore, $\forall$ element in the codomain, there is a corresponding pre-image in the domain, proving that the map r is **surjective**.

Injectivity (One-to-One)

We can establish injectivity by counting the elements (since the sets are finite):

1. **Count the Domain:** The number of elements in the domain is the modulus:

$$|\mathbb{Z}/(mn)\mathbb{Z}| = mn$$

2. **Count the Codomain:** The number of elements in the codomain (a Cartesian product) is the product of the sizes of the factors:

$$|(\mathbb{Z}/m\mathbb{Z}) \times (\mathbb{Z}/n\mathbb{Z})| = |\mathbb{Z}/m\mathbb{Z}| \cdot |\mathbb{Z}/n\mathbb{Z}| = m \cdot n = mn$$

3. **Conclusion:** Since the domain and codomain are finite sets and have the **same number of elements** ($mn = mn$), and we have already proven the map r is **surjective**, it must automatically be **injective** as well.

$$\text{Surjectivity} \wedge (|\text{Domain}| = |\text{Codomain}|) \implies \text{Injectivity}$$

Since r is both surjective and injective, it is a **bijection**. □

Th 9.2: Totient of a Prime Number. *The Totient of a Prime Number: If $m \in \mathbb{N}$ so that m is prime, then $\varphi(m) = m - 1$.*

Th 9.3: Primality Test via the Totient Function. *A Primality Test via the Totient Function: $\forall m \in \mathbb{N} : m \geq 2$, $\varphi(m) = m - 1 \iff m$ is a prime number.*

Proof.

($\Rightarrow$) **("Only if" direction)**: We prove the contrapositive. Assume m is composite (not prime). $\implies \exists p \in \mathbb{N} : 1 < p < m$ and $p \mid m$. Since $p \mid m$, $\gcd(p, m) = p > 1$. Thus, p is not coprime to m. Since $p \in \{1, \dots, m-1\}$ is excluded from the count, the total count $\varphi(m)$ must be strictly less than $m - 1$. More precisely, $\varphi(m) \leq m - 2$. This shows that the equality $\varphi(m) = m - 1$ holds $\implies m$ is prime.

($\Leftarrow$) **("If" direction)**: Assume m is prime. Then $\forall a \in \{1, \dots, m-1\}$, $\gcd(a, m) = 1$. Since there are $m - 1$ such integers, $\varphi(m) = m - 1$. □

Prop 9.2: Totient of a Prime Power. *$\forall p$ prime and $\forall \alpha \in \mathbb{N} : \alpha \geq 1$:*

$$\varphi(p^\alpha) = p^\alpha - p^{\alpha-1}$$

Proof. The totient function $\varphi(p^\alpha)$ counts the number of integers $a \in \{1, \dots, p^\alpha\}$ so that $\gcd(a, p^\alpha) = 1$. The integers that are **not** coprime to p^α are those a for which $p \mid a$. We count these integers (the complement). These non-coprime integers are the multiples of p in the range $[1, p^\alpha]$, i.e., of the form $k \cdot p$ where $k \in \mathbb{N}$. The condition $1 \leq k \cdot p \leq p^\alpha$ implies $1 \leq k \leq p^{\alpha-1}$. Thus, there are exactly $p^{\alpha-1}$ multiples of p in the specified range. Subtracting this count from the total number of integers, p^α, yields:

$$\varphi(p^\alpha) = p^\alpha - p^{\alpha-1} = p^{\alpha-1}(p - 1)$$

□

Th 9.4: Multiplicativity of Euler's Totient Function. *Multiplicativity of Euler's Totient Function: If $m, n \in \mathbb{N}$ are coprime integers ($\gcd(m, n) = 1$), then $\varphi(mn) = \varphi(m)\varphi(n)$.*

10 The fast powering algorithm

Th 10.1: Solvability of a System of Two Linear Congruences. *Let m and n be positive integers, and let a and b be any integers. The system of linear congruences:*

$$\begin{cases} x \equiv a \pmod{m} \\ x \equiv b \pmod{n} \end{cases}$$

has a solution if and only if $\gcd(m, n) \mid (a - b)$.

1: Constructive Proof by Cases (Alessandra's Method). We prove the biconditional nature by demonstrating necessity ($\Longrightarrow$) and sufficiency ($\Longleftarrow$) separately.

1. Necessity ($\Longrightarrow$): We assume a solution exists $\to \exists x_0 \in \mathbb{Z} : x_0 \equiv a \pmod{m}$ and $x_0 \equiv b \pmod{n}$. This implies:

$$x_0 - a = mk \quad \text{for some } k \in \mathbb{Z}$$

$$x_0 - b = nl \quad \text{for some } l \in \mathbb{Z}$$

Thus, $x_0 = a+mk$ and $x_0 = b+nl$. Equating the two expressions for x_0 yields $a+mk = b+nl$. Rearranging the terms: $a - b = nl - mk$. Let $d = \gcd(m, n)$. By definition, $d \mid m$ and $d \mid n$. Therefore, d must divide any integer linear combination of m and n, $\forall k, l \in \mathbb{Z}$. Since $a - b = nl - mk$, we conclude that $d \mid (a - b)$.

2. Sufficiency ($\Longleftarrow$): We assume $\gcd(m, n) \mid (a - b)$. Let $d = \gcd(m, n)$. Since $d \mid (a - b)$, we have $a - b = kd$ for some integer $k \in \mathbb{Z}$. By Bezout's identity, $\exists u, v \in \mathbb{Z} : d = mu + nv$. Substituting the expression for d: $a - b = k(mu + nv) \to a - b = kmu + knv$. Rearranging the equation: $a - kmu = b + knv$. Let $x_0 = a - kmu$. By construction: $x_0 = a - kmu \to x_0 - a = (-ku)m$. Since $-ku \in \mathbb{Z}$, $x_0 \equiv a \pmod{m}$. Also, $x_0 = b+knv \to x_0 - b = (kv)n$. Since $kv \in \mathbb{Z}$, $x_0 \equiv b \pmod{n}$. Since x_0 satisfies both congruences, a solution exists. $\square$

2: Direct Proof via Reduction (Professor's Method). The first congruence $x \equiv a \pmod{m}$ can be rewritten as $x = a+km$ for some integer $k \in \mathbb{Z}$. Substituting this into the second congruence yields:

$$a + km \equiv b \pmod{n}$$

Rearranging the terms, we obtain a single linear congruence for the unknown k:

$$km \equiv b - a \pmod{n}$$

A solution for this linear congruence exists if and only if $\gcd(m, n) \mid (b-a)$. Since $\gcd(m, n) = \gcd(m, n)$ and $\gcd(m, n) \mid (b - a)$ is equivalent to $\gcd(m, n) \mid (a - b)$, the original system has a solution if and only if the condition holds. $\square$

Prop 10.1: General Solvability Condition. *A system of congruences $x \equiv a_i \pmod{m_i}$ for $i \in \{1, 2, \ldots, n\}$ has a solution if and only if for every pair of indices $i \neq j$, the condition $\gcd(m_i, m_j) \mid (a_j - a_i)$ holds.*

A Method to Compute $g^A \pmod N$

Alg 10.1: Fast Powering Algorithm. *Let $g, A, N \in \mathbb{N}$ such that $g \geq 1$, $A \geq 1$, and $N \geq 1$.*

Let $A = (A_r A_{r-1} \dots A_0)_2$ be the binary representation of A, so that $A = A_0 + 2A_1 + 2^2 A_2 + \cdots + 2^r A_r$, where $A_i \in \{0,1\}$ for $0 \leq i \leq r$, and $A_r = 1$.

1. ***Compute the powers** $g^{2^i} \pmod N$ **for** $0 \leq i \leq r$ **by successive squaring:***

$$\begin{aligned} a_0 &\equiv g^{2^0} \equiv g \pmod N \\ a_1 &\equiv a_0^2 \equiv g^{2^1} \pmod N \\ a_2 &\equiv a_1^2 \equiv g^{2^2} \pmod N \\ &\vdots \\ a_r &\equiv a_{r-1}^2 \equiv g^{2^r} \pmod N \end{aligned}$$

   This step requires r multiplications. The next one is the square of the previous one.

2. ***Compute** $g^A \pmod N$ **using the formula** $g^A = g^{A_0 + 2A_1 + 2^2A_2 + \cdots + 2^r A_r}$**:***

$$\begin{aligned} g^A &\equiv g^{A_0} \cdot (g^2)^{A_1} \cdot (g^{2^2})^{A_2} \cdots (g^{2^r})^{A_r} \pmod N \\ &\equiv a_0^{A_0} \cdot a_1^{A_1} \cdot a_2^{A_2} \cdots a_r^{A_r} \pmod N \end{aligned}$$

   This step requires at most r multiplications (since $A_i \in \{0,1\}$).

In total (once the binary form of A is known), we need $2r \approx 2\log_2 A$ steps instead of A (naive method)!

Number of Operations

Rem: Complexity of the Fast Powering Algorithm. *If $2^r \leq A < 2^{r+1}$, then r is the number of bits in the binary representation of A ($\approx \log_2 A$).*

1. *It requires at most r divisions to write A in binary ($\sim r$ bits).*
2. *To compute g^A, where $A = A_0 + 2A_1 + 2^2A_2 + \cdots + 2^r A_r$, we need:*
   - *r multiplications for the successive squaring (Step 1).*
   - *At most r multiplications for the final product (Step 2).*

   Thus, we need $\sim 2r$ multiplications to compute $g^A \pmod N$.

   Since $2^r \leq A$, it follows that $r \leq \log_2(A)$. Therefore, the total number of multiplications is $\leq 2r \leq 2\log_2(A)$.

Prop 10.2: Efficiency Comparison. *If $A \approx 2^{1000} \approx 10^{300}$, then $\log_2(A) \approx 1000$. The Fast Powering Algorithm requires $\sim 2\log_2(A) \approx 2000$ multiplications.*

In contrast, the Naive Method requires $A \approx 10^{300}$ multiplications. The Fast Powering Algorithm is dramatically more efficient.

Rem: Problem Statement. *The algorithm is designed to efficiently calculate the value of $g^A \pmod N$, given integers $g > 1, A > 1$, and $N > 1$. The challenge arises when the exponent A is very large, $\forall A \in \mathbb{N} : A \gg 1$. The naive method requires $\mathcal{O}(A)$ multiplications, which is computationally infeasible for large A.*

Rem: Method Principle. *The Fast Powering Algorithm leverages the binary representation of the exponent A. Let the binary expansion be $A = \sum_{i=0}^{r} A_i 2^i$, where $A_i \in \{0, 1\}$. The exponentiation property allows us to write:*

$$g^A = g^{\sum_{i=0}^{r} A_i 2^i} = \prod_{i=0}^{r} (g^{2^i})^{A_i} \pmod N$$

*The core idea is **Successive Squaring**: the terms $g^{2^i} \pmod N$ are computed efficiently by starting with g^1 and repeatedly squaring the previous result, $\forall i \in \{1, \ldots, r\} : g^{2^i} = (g^{2^{i-1}})^2 \pmod N$. The final result is the product of only those terms $g^{2^i} \pmod N$ for which the corresponding binary coefficient $A_i = 1$.*

Rem: Complexity Analysis. *The Fast Powering Algorithm significantly reduces complexity from $\mathcal{O}(A)$ (naive method) to $\mathcal{O}(\log_2(A))$. This is due to the three phases:*

- *Binary Conversion: $\sim \log_2(A)$ divisions.*
- *Successive Squaring: $\sim \log_2(A)$ squarings.*
- *Final Product: at most $\sim \log_2(A)$ multiplications.*

The total number of operations $\sim 3\log_2(A)$, providing immense efficiency gain, $\forall A \in \mathbb{N} : A \approx 2^{1000} \to$ Operations ≈ 3000.

Th 10.2: Fermat's Little Theorem - Primary Form. *Let p be a prime number. For any integer $a \in \mathbb{Z}$ so that p does not divide a, we have:*

$$a^{p-1} \equiv 1 \pmod p$$

Th 10.3: Fermat's Little Theorem - Alternative Form. *Let p be a prime number. For any integer $a \in \mathbb{Z}$, we have:*

$$a^p \equiv a \pmod{p}$$

Rem: Conditions for Application. *The theorem is generally false for composite moduli. The primary condition is that the modulus p must be a prime number.*

11 Fermat's little theorem

Th 11.1: Fermat's Little Theorem. *Let p be a prime. If $a \in (\mathbb{Z}/p\mathbb{Z})^*$, then $a^{p-1} = 1$. Equivalently, if p is a prime and a is an integer so that p does not divide a, then $a^{p-1} \equiv 1 \pmod{p}$.*

Proof. The proof employs a technique involving the set of invertible elements modulo p.

Step 1: Construct the Set Let $a \in (\mathbb{Z}/p\mathbb{Z})^*$. Consider the set S formed by multiplying a by each of the $p-1$ invertible elements of $\mathbb{Z}/p\mathbb{Z}$, which are $I = \{1, 2, \ldots, p-1\}$.

$$S = \{a \cdot i \pmod{p} \mid \forall i \in I\}$$

Step 2: Prove Non-Zero Property We claim that $a \cdot i \not\equiv 0 \pmod{p}$ for all $i \in I$. Assume for contradiction that $a \cdot i \equiv 0 \pmod{p}$. This implies $p \mid (a \cdot i)$. Since p is prime, $p \mid a$ or $p \mid i$. By premise, $a \in (\mathbb{Z}/p\mathbb{Z})^*$, so $p \nmid a$. Since $1 \leq i \leq p-1$, $\gcd(i, p) = 1$, so $p \nmid i$. This is a contradiction, so $\forall i \in I$, $a \cdot i \not\equiv 0 \pmod{p}$.

Step 3: Prove Distinctness We claim that all $p-1$ elements in S are distinct. Assume for contradiction that $\exists i, j \in I$ so that $i \neq j$ and $a \cdot i \equiv a \cdot j \pmod{p}$. This implies $a \cdot (i-j) \equiv 0 \pmod{p}$. Since $p \nmid a$, we must have $p \mid (i-j)$. However, since $i, j \in \{1, \ldots, p-1\}$ and $i \neq j$, the difference $i-j$ satisfies $0 < |i-j| < p$. Such an integer cannot be a multiple of p. This contradiction proves that all elements in S are distinct.

Step 4: Equate Sets Since S contains $p-1$ distinct, non-zero elements modulo p, and the set of all non-zero elements is $(\mathbb{Z}/p\mathbb{Z})^* = I = \{1, 2, \ldots, p-1\}$, the two sets must be identical, $\forall i \in I : S = I$.

Step 5: The Trick (Multiplication) The product of the elements in S must equal the product of the elements in I:

$$\prod_{i=1}^{p-1}(a \cdot i) \equiv \prod_{i=1}^{p-1} i \pmod{p}$$

Step 6: Simplify and Conclude Factoring the left-hand side yields:

$$a^{p-1} \cdot (p-1)! \equiv (p-1)! \pmod{p}$$

Since p is prime, $(p-1)!$ is invertible modulo p (by Wilson's Theorem, or simply because $\forall i \in \{1, \ldots, p-1\}$, $\gcd(i, p) = 1$). We can safely cancel $(p-1)!$ from both sides:

$$a^{p-1} \equiv 1 \pmod{p}$$

□

Cor 11.1: Modular Exponentiation. *Let p be a prime, and $a \in \mathbb{Z}$. If $\forall x, y \in \mathbb{N} \setminus \{0\}$ so that $x \equiv y \pmod{p-1}$, then $a^x \equiv a^y \pmod{p}$.*

Proof. We consider two distinct cases for the integer a.

Case 1: p divides a If $p \mid a$, then $a \equiv 0 \pmod p$. Since $x \geq 1$ and $y \geq 1$, we have $a^x \equiv 0 \pmod p$ and $a^y \equiv 0 \pmod p$. Thus, $a^x \equiv a^y \pmod p$ holds trivially.

Case 2: p does not divide a The congruence $x \equiv y \pmod{p-1}$ means $y = x + k(p-1)$ for some $k \in \mathbb{Z}$. Substituting this into a^y:

$$a^y = a^{x+k(p-1)} = a^x \cdot a^{k(p-1)} = a^x \cdot (a^{p-1})^k$$

By Fermat's Little Theorem, $a^{p-1} \equiv 1 \pmod p$. Therefore:

$$a^y \equiv a^x \cdot (1)^k \pmod p \rightarrow a^y \equiv a^x \pmod p$$

□

Cor 11.2: Finding Modular Inverses. *Let p be a prime and $a \in (\mathbb{Z}/p\mathbb{Z})^*$. Then the inverse of a modulo p is a^{p-2}.*

Proof. We must show that $a \cdot a^{p-2} \equiv 1 \pmod p$.

$$a \cdot a^{p-2} = a^{1+p-2} = a^{p-1}$$

By Fermat's Little Theorem, $\forall a \in \mathbb{Z}$ so that $p \nmid a$, we have $a^{p-1} \equiv 1 \pmod p$. Therefore:

$$a \cdot a^{p-2} \equiv 1 \pmod p$$

This confirms that a^{p-2} is the multiplicative inverse of a modulo p. □

Cor 11.3: Fermat Primality Test. *Let p be an integer so that $p \geq 2$, and let a be an integer so that p does not divide a. If $a^{p-1} \not\equiv 1 \pmod p$, then p is not a prime number.*

Proof. This statement is the contrapositive of Fermat's Little Theorem: $(\neg\text{THEN}) \implies (\neg\text{IF})$. The original theorem states: IF p is prime, THEN $a^{p-1} \equiv 1 \pmod p$. The contrapositive is: IF $a^{p-1} \not\equiv 1 \pmod p$, THEN p is not prime. This conclusion is logically sound. □

12 The Order of an Element and Primitive Roots

Def 12.1: Order of an Element. *Let p be a prime and $a \in \mathbb{Z}$ so that p does not divide a. The order of a modulo p, denoted $\text{ord}_p(a)$, is the minimum positive integer $n \in \mathbb{N} \setminus \{0\}$ so that $a^n \equiv 1 \pmod{p}$.*

Prop 12.1: Order Divisibility Property. *Let p be a prime, $a \in \mathbb{Z}$ so that p does not divide a, and $n \geq 1$. Then $a^n \equiv 1 \pmod{p}$ if and only if $\text{ord}_p(a)$ divides n.*

Proof. We demonstrate both directions of the biconditional.

First Direction ($\Longleftarrow$): Assume $\text{ord}_p(a) \mid n$. $\rightarrow \exists k \in \mathbb{Z}$ so that $n = k \cdot \text{ord}_p(a)$. Then:

$$a^n = a^{k \cdot \text{ord}_p(a)} = (a^{\text{ord}_p(a)})^k$$

By definition of the order, $a^{\text{ord}_p(a)} \equiv 1 \pmod{p}$. Therefore:

$$a^n \equiv (1)^k \equiv 1 \pmod{p}$$

Second Direction ($\Longrightarrow$): Assume $a^n \equiv 1 \pmod{p}$. By the division algorithm, $\exists q, r \in \mathbb{Z}$ so that $n = q \cdot \text{ord}_p(a) + r$, where $0 \leq r < \text{ord}_p(a)$. We evaluate a^n:

$$a^n = a^{q \cdot \text{ord}_p(a) + r} = (a^{\text{ord}_p(a)})^q \cdot a^r$$

Substituting the assumed and known congruences:

$$1 \equiv (1)^q \cdot a^r \pmod{p} \rightarrow a^r \equiv 1 \pmod{p}$$

Since $\text{ord}_p(a)$ is the *minimum* positive integer for which this holds, and $0 \leq r < \text{ord}_p(a)$, the only possibility is that $r = 0$. $\rightarrow \text{ord}_p(a)$ divides n. □

Prop 12.2: Equality of Powers. *Let p be a prime and $g \in (\mathbb{Z}/p\mathbb{Z})^*$. Then $g^x = g^y$ if and only if $x \equiv y \pmod{\text{ord}_p(g)}$.*

Proof. Since $g \in (\mathbb{Z}/p\mathbb{Z})^*$, g is invertible.

$$g^x = g^y \iff g^x \cdot g^{-y} = 1 \iff g^{x-y} = 1$$

Applying the proposition from the previous section (2.1), $g^n = 1$ if and only if $\text{ord}_p(g) \mid n$. Let $n = x - y$. We conclude:

$$g^{x-y} = 1 \iff \text{ord}_p(g) \mid (x - y)$$

By the definition of modular congruence, this is equivalent to: $x \equiv y \pmod{\text{ord}_p(g)}$. □

Def 12.2: Primitive Root. *An element $g \in (\mathbb{Z}/p\mathbb{Z})^*$ is a primitive root if its powers generate all the elements of $(\mathbb{Z}/p\mathbb{Z})^*$. That is, the set $\{g^1, g^2, \ldots, g^{p-1}\}$ is equal to the set $\{1, 2, \ldots, p-1\}$.*

Th 12.1: Primitive Root Condition. *An element $g \in (\mathbb{Z}/p\mathbb{Z})^*$ is a primitive root if and only if $ord_p(g) = p - 1$.*

Proof Outline. For g to be a primitive root, the $p - 1$ powers $\{g^1, g^2, \ldots, g^{p-1}\}$ must all be distinct. By the Proposition on the Condition for Equality of Powers, $g^i = g^j$ if and only if $i \equiv j \pmod{\mathrm{ord}_p(g)}$. For the powers to be distinct for all $1 \le i < j \le p-1$, we must ensure that $g^{j-i} \neq 1$ for any exponent $j - i$ where $1 \le j - i < p - 1$. This means that the smallest positive exponent k so that $g^k = 1$ must be $p - 1$. By definition, $\mathrm{ord}_p(g) = k$. Therefore, g is a primitive root if and only if $\mathrm{ord}_p(g) = p - 1$. □

Def 12.3. *Let p be a prime, and $a \in \mathbb{Z}$ such that $p \nmid a$. The* ***order*** *$ord_p(a)$ of a modulo p is the minimum $n \in \mathbb{N}$ such that $n \geq 1$ and $a^n \equiv 1 \pmod{p}$.*

Prop 12.3. *Let p be a prime, $a \in \mathbb{Z}$ such that $p \nmid a$, and $n \geq 1$. Then $a^n \equiv 1 \pmod{p}$ if and only if the order of a modulo p, denoted $ord_p(a)$, divides n. In particular, $ord_p(a) \mid (p-1)$ (a consequence of Fermat's Little Theorem).*

Proof. Let $k = \mathrm{ord}_p(a)$.

($\Rightarrow$) **If $a^n \equiv 1 \pmod{p}$, then $k \mid n$**
Use the Division Algorithm on n by k: $n = qk + r$, where $0 \le r < k$.

$$1 \equiv a^n \equiv a^{qk+r} \equiv (a^k)^q \cdot a^r \equiv 1^q \cdot a^r \equiv a^r \pmod{p}$$

Since $a^r \equiv 1 \pmod{p}$ and k is the **minimum positive exponent** satisfying this property, and $r < k$, we must have $r = 0$. Thus, $n = qk$, which means $k \mid n$.

($\Leftarrow$) **If $k \mid n$, then $a^n \equiv 1 \pmod{p}$**
If $k \mid n$, then $n = qk$ for some $q \in \mathbb{Z}$.

$$a^n = a^{qk} = (a^k)^q \equiv 1^q \equiv 1 \pmod{p}$$

□

Cor 12.1. *Let p be a prime, $a \in \mathbb{Z}$ such that $p \nmid a$. $ord_p(a) = \min\{n \in \mathbb{N} : n \geq 1$ so that $a^n \equiv 1 \pmod{p}\} = \min\{n \in \mathbb{N} : n \geq 1$ so that $a^n \equiv 1 \pmod{p}$ and $n \mid p-1\}$.*

Prop 12.4. *Let p be a prime. If $g \in \mathbb{F}_p^*$, then $g^x = g^y \iff y \equiv x \pmod{ord_p(g)}$.*

Proof. ($\Leftarrow$) Assume $x \equiv y \pmod{ord_p(g)}$. Then $x - y = k \cdot ord_p(g)$ for some integer k. $g^x = g^{y+k \cdot ord_p(g)} = g^y \cdot (g^{ord_p(g)})^k \equiv g^y \cdot 1^k \equiv g^y \pmod{p}$.

($\Rightarrow$) Assume $g^x = g^y$. Without loss of generality, assume $x \geq y$. Since $g \in \mathbb{F}_p^*$ is invertible, multiplying by $(g^{-1})^y$ gives $g^{x-y} = 1$. By the previous Proposition, $ord_p(g) \mid x - y$, which means $x \equiv y \pmod{ord_p(g)}$. □

Def 12.4. *Let p be a prime. A* ***primitive root*** *of $\mathbb{F}_p$ is any $g \in \mathbb{F}_p^*$ such that the powers of g generate all the elements of $\mathbb{F}_p^*$, i.e., $\mathbb{F}_p^* = \{1, g, g^2, \ldots, g^{p-2}\}$.*

Th 12.2: Characterization of the Primitive Roots. *Let p be a prime. $g \in \mathbb{F}_p^*$ is a primitive root if and only if $ord_p(g) = p - 1$.*

Proof. The proof establishes the equivalence between g generating the group $\mathbb{F}_p^*$ and g having the maximum possible order. Note that $|\mathbb{F}_p^*| = p - 1$.

($\Rightarrow$) If g is a primitive root, then ord$_p(g) = p - 1$
By definition, if g is a primitive root, the set of its powers $\{1, g, \ldots, g^{p-2}\}$ contains **all** $p - 1$ **distinct elements** of $\mathbb{F}_p^*$.

Let $m = \text{ord}_p(g)$. If we assume $m < p - 1$, then the sequence of powers $1, g, g^2, \ldots$ repeats with a period of m. This means the set of distinct powers $\langle g \rangle$ has size $|\langle g \rangle| = m$.

Since g is a primitive root, $\langle g \rangle = \mathbb{F}_p^*$, so $m = |\mathbb{F}_p^*| = p - 1$. The assumption $m < p - 1$ leads to the contradiction $|\langle g \rangle| < |\mathbb{F}_p^*|$. Thus, $m = p - 1$.

($\Leftarrow$) If ord$_p(g) = p - 1$, then g is a primitive root
Assume $\text{ord}_p(g) = p - 1$. We must show the powers $1, g, \ldots, g^{p-2}$ are all distinct.

Suppose, for contradiction, that $g^i \equiv g^j \pmod{p}$ for $0 \leq i < j \leq p - 2$. Since g is invertible, $g^{j-i} \equiv 1 \pmod{p}$.

Let $k = j - i$. Then $0 < k < p - 1$. This implies that a power $g^k \equiv 1 \pmod{p}$ for an exponent k smaller than $p - 1$. This contradicts the assumption that $\text{ord}_p(g) = p - 1$ is the **minimum** positive exponent that results in 1.

Thus, the $p - 1$ powers $\{1, g, \ldots, g^{p-2}\}$ are distinct. Since $\mathbb{F}_p^*$ has exactly $p - 1$ elements, the powers must generate the entire group, meaning g is a primitive root. □

Th 12.3: Existence of Primitive Roots. *Let p be a prime. $\mathbb{F}_p$ has at least one primitive root. (Admitted)*

Proof. Let $x = g^{\frac{p-1}{2}}$. We first observe the square of x:

$$x^2 = \left(g^{\frac{p-1}{2}}\right)^2 = g^{p-1}$$

By **Fermat's Little Theorem**, $g^{p-1} \equiv 1 \pmod p$, provided $p \nmid g$ (which is true since $g \in \mathbb{F}_p^*$). Thus, x satisfies the congruence:

$$x^2 \equiv 1 \pmod p$$

In the field $\mathbb{F}_p$, the only solutions to $x^2 \equiv 1$ are $x = 1$ and $x = -1$ (or $x \equiv p-1 \pmod p$).

We must show that $x \neq 1$. We proceed by contradiction.

Assume $x = 1$, meaning:

$$g^{\frac{p-1}{2}} \equiv 1 \pmod p$$

By the property of the order of an element, this would imply:

$$\text{ord}_p(g) \mid \frac{p-1}{2}$$

However, since g is a **primitive root**, we know by definition that $\text{ord}_p(g) = p-1$. The divisibility condition becomes:

$$(p-1) \mid \frac{p-1}{2}$$

Since $p > 2$, this condition is false (a positive integer cannot divide a number half its size). This is a **contradiction**.

Therefore, the only remaining possibility for the value of x is $x = -1$.

$$g^{\frac{p-1}{2}} \equiv -1 \pmod p$$

□

Rem. *Let $p > 2$ be a prime. If g is a square in $\mathbb{F}_p$, then g is not a primitive root in $\mathbb{F}_p$.*

Proof. Let g be a square, so $g = h^2$ for some $h \in \mathbb{F}_p^*$. Then $g^{(p-1)/2} = (h^2)^{(p-1)/2} = h^{p-1} \equiv 1 \pmod p$ by Fermat's Little Theorem. Since $g^{(p-1)/2} \equiv 1 \pmod p$, by the proposition on order, $ord_p(g) \mid \frac{p-1}{2}$. Since $\frac{p-1}{2} < p-1$ (as $p > 2$), we have $ord_p(g) < p-1$. By the characterization theorem, g is not a primitive root. □

Lemma 12.1: Effective Test for a Primitive Root Candidate. *Let $p > 2$ be prime and let $p - 1 = \prod_{i=1}^{m} p_i^{a_i}$ be the prime factorization of $p-1$. Then $g \in \mathbb{F}_p^*$ is a primitive root if and only if $g^{(p-1)/p_i} \not\equiv 1 \pmod p$ for all $i = 1, \ldots, m$.*

Proof. Let $k = \text{ord}_p(g)$. Recall that g is a primitive root $\iff k = p - 1$.

($\Rightarrow$) **If g is a primitive root, the test passes**
Assume g is a primitive root, so $k = p - 1$.

Suppose, for contradiction, that $g^{(p-1)/p_i} \equiv 1 \pmod p$ for some prime factor p_i. By the property of the order, k must divide the exponent:

$$k \mid \frac{p-1}{p_i} \implies (p-1) \mid \frac{p-1}{p_i}$$

Since p_i is a prime ($p_i \geq 2$), this divisibility is impossible. Thus, $g^{(p-1)/p_i} \not\equiv 1 \pmod p$ must hold for all i.

($\Leftarrow$) **If the test passes, g is a primitive root**
Assume $g^{(p-1)/p_i} \not\equiv 1 \pmod p$ for all i. We know $k \mid p - 1$, so k is a divisor of $p - 1$, and thus k must be of the form $k = p_1^{f_1} \cdots p_m^{f_m}$ with $0 \leq f_i \leq a_i$.

Suppose, for contradiction, that $k < p - 1$. This means at least one exponent f_i must be strictly less than a_i (i.e., $f_i \leq a_i - 1$).

For such an index i, we can construct the exponent $\frac{p-1}{p_i}$. Since $p_i^{a_i}$ is a factor of $p - 1$ and $p_i^{f_i}$ is a factor of k, the difference in the exponent is:

$$\frac{p-1}{p_i} = \frac{p_1^{a_1} \cdots p_i^{a_i - 1} \cdots p_m^{a_m}}{p_i}$$

Since $f_i \leq a_i - 1$, we have $k \mid \frac{p-1}{p_i}$.

By the property of the order, $k \mid \frac{p-1}{p_i}$ implies $g^{(p-1)/p_i} \equiv 1 \pmod p$. This contradicts our initial assumption that $g^{(p-1)/p_i} \not\equiv 1 \pmod p$ for all i.

Therefore, the assumption $k < p - 1$ is false, meaning $k = p - 1$. Hence, g is a primitive root. $\square$

Alg 12.1: Finding a Primitive Root. *Let $p > 2$ be prime and $p - 1 = \prod_{i=1}^{m} p_i^{a_i}$ be the prime factorization of $p - 1$, with p_i distinct primes and $a_i \geq 1$ for $i = 1, \ldots, m$.*

1. *Start with $g = 2$.*
2. *If $g^{(p-1)/p_i} \not\equiv 1 \pmod p$ for all $i = 1, \ldots, m$, then g is a primitive root.*
3. *If not, increment $g \to g + 1$ and return to Step 2.*

13 Number of Primitive Roots

Th 13.1: Characterization of Powers as Primitive Roots. *Let p be a prime and g be a primitive root of $\mathbb{F}_p$. For any $i \in \{1, \dots, p-2\}$, we have:*

1. *g^i is a primitive root of $\mathbb{F}_p \iff \gcd(i, p-1) = 1$.*
2. *$\mathbb{F}_p$ has exactly $\phi(p-1)$ primitive roots.*

Proof of 1. Let $p - 1 = p_1^{a_1} \cdots p_m^{a_m}$ with p_j distinct primes and $a_j \geq 1$. By the effective test for a primitive root (Lemma from Lesson 5), g^i is not a primitive root $\iff (g^i)^{(p-1)/p_j} \equiv 1 \pmod p$ for some j. We analyze the condition $(g^i)^{(p-1)/p_j} \equiv 1 \pmod p$:

$$(g^i)^{(p-1)/p_j} \equiv 1 \pmod p \iff ord_p(g) \mid \frac{i(p-1)}{p_j} \quad \text{since } g \text{ is a primitive root, } ord_p(g) = p-1$$

$$\iff p - 1 \mid \frac{i(p-1)}{p_j}$$

$$\iff p_j \mid i$$

Thus, g^i is not a primitive root $\iff p_j \mid i$ for some j. Since the p_j are the distinct prime factors of $p-1$, the condition "$p_j \mid i$ for some j" is equivalent to $\gcd(i, p-1) > 1$. Therefore, g^i is a primitive root $\iff \gcd(i, p-1) = 1$. □

Proof of 2. Since g is a primitive root, the elements of $\mathbb{F}_p^*$ are given by the distinct powers $\{1, g, \dots, g^{p-2}\}$. From part 1, the other primitive roots are the powers g^i such that $1 \leq i \leq p-2$ and $\gcd(i, p-1) = 1$. The number of such integers i in the range $\{1, \dots, p-2\}$ (or $\{1, \dots, p-1\}$) that are relatively prime to $p-1$ is exactly $\phi(p-1)$, where ϕ is Euler's totient function. Therefore, $\mathbb{F}_p$ has $\phi(p-1)$ primitive roots. □

Lemma 13.1. *Let $a, b, c \in \mathbb{Z}$ with $a \neq 0$. Then $a \mid bc \iff \frac{a}{\gcd(a,b)} \mid c$.*

Proof. Let $d = \gcd(a, b)$. We set $a = a'd$ and $b = b'd$, where $\gcd(a', b') = 1$. The statement $a \mid bc$ is equivalent to $a'd \mid b'dc$. Since $d \neq 0$, we can cancel d: $a' \mid b'c$. Because $\gcd(a', b') = 1$, by Euclid's Lemma, $a' \mid b'c \iff a' \mid c$. Substituting back $a' = \frac{a}{d} = \frac{a}{\gcd(a,b)}$, we get the equivalence: $a \mid bc \iff \frac{a}{\gcd(a,b)} \mid c$. □

Prop 13.1: The Order of a Power. *Let p be a prime, g be a primitive root in $\mathbb{F}_p$, and $i \in \{1, \dots, p-2\}$. Then $ord_p(g^i) = \frac{p-1}{\gcd(p-1,i)}$.*

Proof. The order $ord_p(g^i)$ is defined as $m = \min\{n \in \mathbb{N} : n \geq 1 \text{ so that } (g^i)^n \equiv 1 \pmod p\}$. The congruence $(g^i)^n \equiv 1 \pmod p$, or $g^{in} \equiv 1 \pmod p$, is equivalent to $ord_p(g) \mid in$. Since g is a primitive root, $ord_p(g) = p - 1$. Thus, we are looking for:

$$ord_p(g^i) = \min\{n \in \mathbb{N} : n \geq 1 \text{ so that } (p-1) \mid in\}$$

Using the preceding Lemma with $a = p-1$, $b = i$, and $c = n$, the condition $(p-1) \mid in$ is equivalent to $\frac{p-1}{\gcd(p-1,i)} \mid n$. The minimum positive integer n such that $\frac{p-1}{\gcd(p-1,i)} \mid n$ is $n = \frac{p-1}{\gcd(p-1,i)}$. Therefore, $ord_p(g^i) = \frac{p-1}{\gcd(p-1,i)}$. $\square$

Cor 13.1: Alternative Proof. *Let g be a primitive root in $\mathbb{F}_p$. Then g^i is a primitive root $\iff \gcd(p-1,i) = 1$.*

Proof. By the Characterization Theorem, g^i is a primitive root $\iff ord_p(g^i) = p-1$. Using the Proposition on the order of a power:

$$\begin{aligned} ord_p(g^i) = p-1 &\iff \frac{p-1}{\gcd(p-1,i)} = p-1 \\ &\iff \gcd(p-1,i) = 1 \end{aligned}$$

$\square$

Artin's Conjecture

Rem. *If $g \in \mathbb{N}$ is a perfect square, there is no prime p such that $p > g$ and g is a primitive root in $\mathbb{F}_p$.*

Th 13.2: Artin's Conjecture (Hooley, 1967 - under GRH). *If $g \geq 2$ is not a perfect square, the set of primes p for which g is a primitive root in $\mathbb{F}_p$ has a positive density.*

$$\lim_{x \to +\infty} \frac{\#\{p \leq x : p \text{ is prime and } g \text{ is a primitive root in } \mathbb{F}_p\}}{\#\{p : p \text{ is prime, } p \leq x\}} = C_g > 0$$

For $g = 2$, the constant is $C_2 \approx 0.3739558\ldots$

Th 13.3: Roger-Heath-Brown, 1985 - unconditional. *There are at most two primes that are not primitive roots for infinitely many primes.*
Rem. *This theorem implies that among the first few primes, like 2, 3, and 5, at least one of them must be a primitive root for infinitely many primes.*